



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

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- Methodology
- Results
- Conclusion
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Executive Summary

Problem Statement

The commercial space transportation industry is driven by cost efficiency, where SpaceX's ability to land and reuse the first stage of the Falcon 9 rocket significantly lowers launch expenses—reducing costs from upwards of \$165 million per launch to roughly \$62 million. Predicting whether a first-stage booster will land successfully is critical for dynamic pricing, launch scheduling, and risk management. This project addresses this core business challenge by predicting first-stage landing success based on historical launch parameters, enabling data-driven operational decision-making.

Methodology & Approach

To solve this problem, a end-to-end data science pipeline was executed using Python, SQL, and Scikit-learn. Historical launch data was collected via SpaceX REST API endpoints and web scraping, then structured and queried using SQL database operations. Interactive exploratory data analysis was performed using Folium maps to analyze geographic proximity to launch sites and Plotly Dash web applications to examine correlations across payload mass, booster version, and launch site locations. Finally, classification machine learning models—including Logistic Regression, Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbors (KNN)—were trained, tuned with Grid Search CV, and evaluated to optimize predictive accuracy.

Key Findings & Business Impact

The analysis revealed that payload mass, orbit type (e.g., LEO vs. GTO), and launch site location are the primary determinants of successful booster recovery. Among the trained algorithms, several classification models achieved a top testing accuracy of 83.3%, demonstrating strong reliability in predicting landing success under hyperparameter tuning. These predictive capabilities allow commercial satellite operators and space logistics planners to assess mission risk accurately, streamline operational pricing, and maximize the financial benefits of booster reusability.

Introduction

Background & Business Context

Over the past decade, SpaceX has fundamentally disrupted the commercial spaceflight industry by introducing reusability into orbital launch systems. Historically, primary rocket boosters were disposable, contributing significantly to high mission costs exceeding \$165 million per launch. By mastering autonomous first-stage landings—both on land landing zones (RTLS) and offshore autonomous spaceport drone ships (ASDS)—SpaceX successfully reduced average launch costs down to approximately \$62 million. Predicting whether a first-stage booster can be successfully recovered allows space agencies, commercial satellite operators, and launch service aggregators to dynamically calculate payload pricing, mitigate insurance risks, and optimize launch availability schedules.

Dataset Sources & Data Pipeline

The analysis leverages multi-source primary data collected directly from spaceflight records:

- * SpaceX REST API: Ingestion of detailed mission metadata, including booster serial numbers, payload masses, target orbit profiles, landing pad locations, and outcome telemetry.
- * Web Scraping: Automated extraction of historical launch logs, payload details, and core reusability histories from Wikipedia launch records.
- * SQL & Data Cleaning Pipeline: Structuring, filtering, and standardizing raw API JSON responses into relational tables for relational SQL querying and exploratory data analysis.

Project Objectives

- * Exploratory & Geographic Analysis: Determine the statistical relationships between payload mass, destination orbit type (e.g., LEO, GTO, ISS, SSO), and booster success rates using custom SQL queries, Seaborn/Matplotlib visualization pipelines, and interactive Folium geospatial maps.
- * Interactive Dashboarding: Build a dynamic Plotly Dash application to allow stakeholders to visually isolate and filter launch outcome probabilities across different payload ranges and launch sites.
- * Machine Learning Modeling: Build, train, and optimize supervised classification algorithms—including Logistic Regression, Support Vector Machines (SVM), Decision Trees, and K-Nearest Neighbors (KNN)—using Scikit-learn and hyperparameter tuning (GridSearchCV) to predict booster landing outcomes on unseen test data.

Scope & Boundaries

The scope of this project is limited to historical commercial and government orbital launches of the Falcon 9 launch vehicle family. It evaluates launch parameters (payload size, orbit classification, launch venue, booster flights) and landing parameters (landing site type, booster failure/success outcomes). Factors such as real-time atmospheric weather conditions, telemetry flight trajectories, and post-landing refurbishing costs fall outside the primary scope of this analysis.

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - API Extraction & Web Scraping: Extracted 90 historical Falcon 9 launch records via the SpaceX REST API and scraped complementary launch outcome details from Wikipedia using BeautifulSoup. Describe how data was collected
- Perform data wrangling
 - Cleaning & Feature Preprocessing: Filtered for Falcon 9 launches, imputed missing PayloadMass values with dataset averages, created a binary target variable Class (1 = \text{Successful Landing}, 0 = \text{Failed Landing}), and applied One-Hot Encoding to categorical variables (LaunchSite, Orbit). Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to Model Training & Evaluation: Standardized feature sets with StandardScaler and trained four supervised classifiers—Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)—tuning hyper-parameters via GridSearchCV with 10-fold cross-validation to prevent overfitting., tune, evaluate classification models

Data Collection

SpaceX REST API Integration:

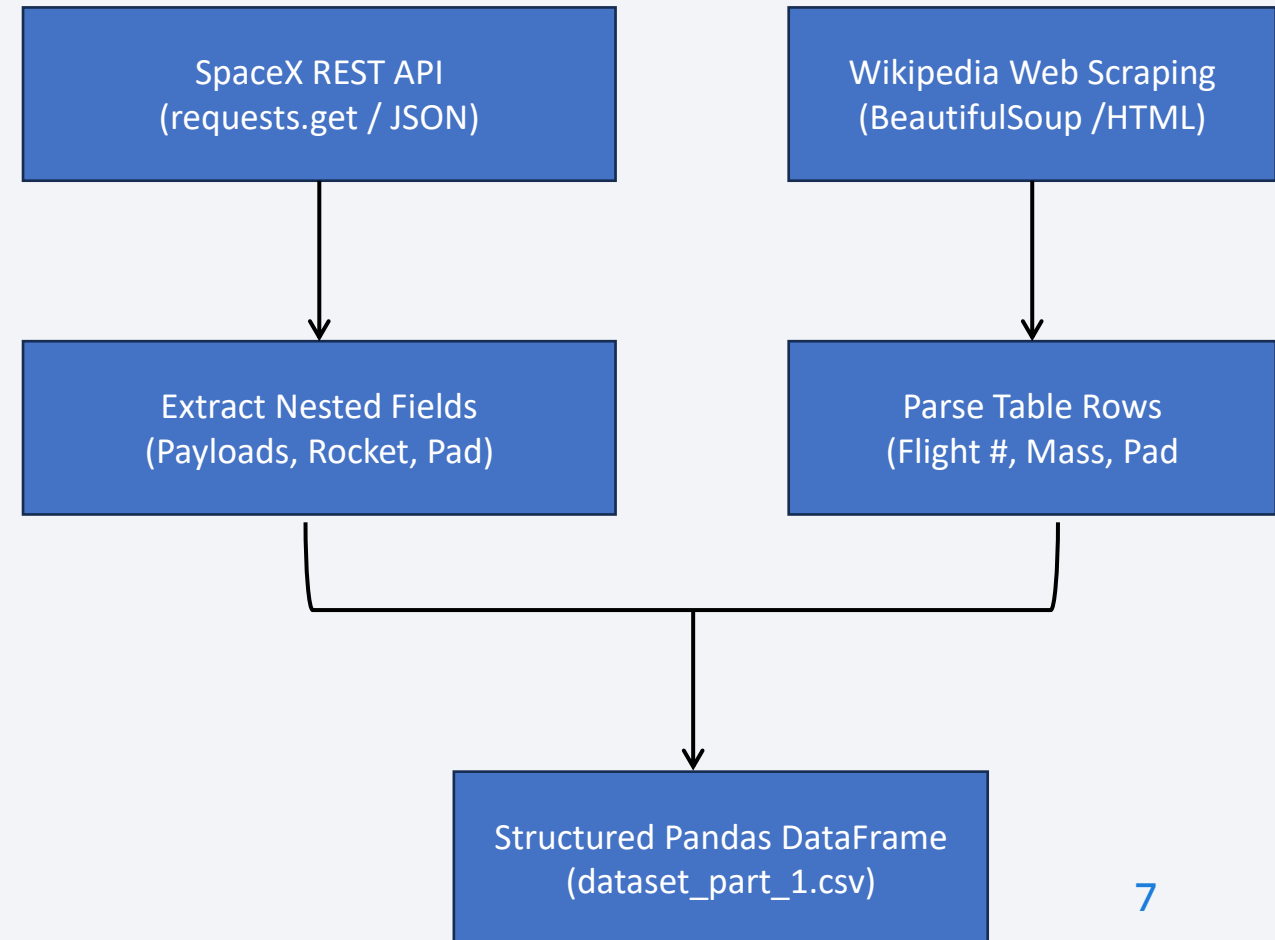
- Called the SpaceX API endpoint (v4/launches/past) to extract historical launch, payload, and rocket specifications.
- Extracted nested JSON fields using custom helper functions (getBoosterVersion, getPayloadData, getLaunchSite, getCoreData).

Web Scraping Wikipedia:

- Scraped Falcon 9 launch records from Wikipedia using BeautifulSoup to parse HTML table data.
- Extracted flight numbers, launch dates, payload masses, orbit types, and landing outcomes.

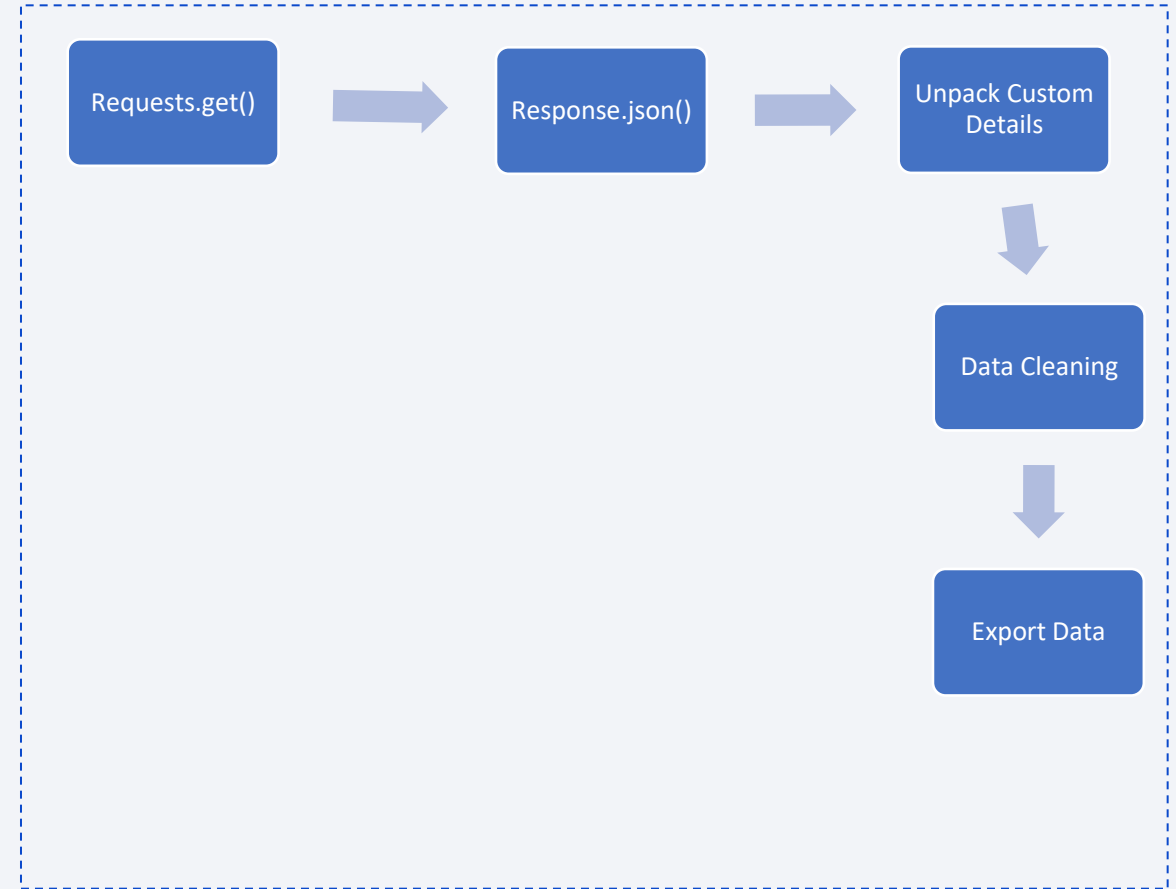
Data Integration:

- Normalized and formatted both datasets into structured Pandas DataFrames for downstream wrangling and modeling.



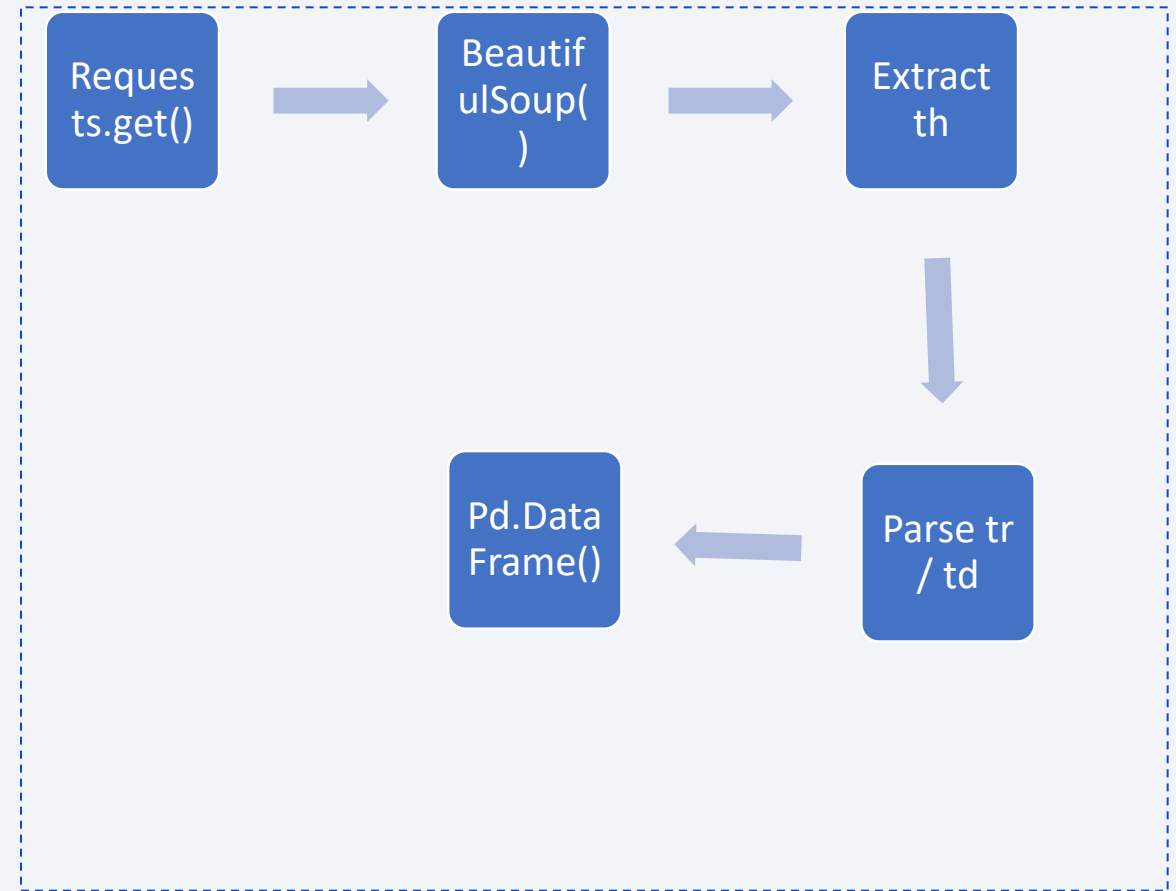
Data Collection – SpaceX API

- REST API Endpoint: Requested launch data from SpaceX API
(https://api.spacexdata.com/v4/launches/past).
- Data Unpacking: Expanded nested JSON responses for launchpads, rockets, payloads, and cores.
- Data Cleaning: Handled missing values (e.g., filled null core payload mass/landings) and formatted date fields.
- Export: Filtered Falcon 9 historical records into a clean Pandas DataFrame and saved as CSV.
- [IBM-Data-Science-Capstone/spacex-dash-app.py at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)



Data Collection - Scraping

- Source URL: Target Wikipedia page for Falcon 9 / Falcon Heavy launches.
- HTTP Request: Send GET request using requests library with custom headers.
- HTML Parsing: Create BeautifulSoup object using html.parser to parse DOM structure.
- Header Extraction: Filter table elements (wikitable) and extract column names from <th> tags.
- Data Parsing: Loop through rows (<tr>/<td>), parse columns, and populate launch_dict.
- DataFrame & Export: Convert structured dictionary to Pandas DataFrame and export to CSV.
- [IBM-Data-Science-Capstone/jupyter-labs-webscraping\(2\).ipynb at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)



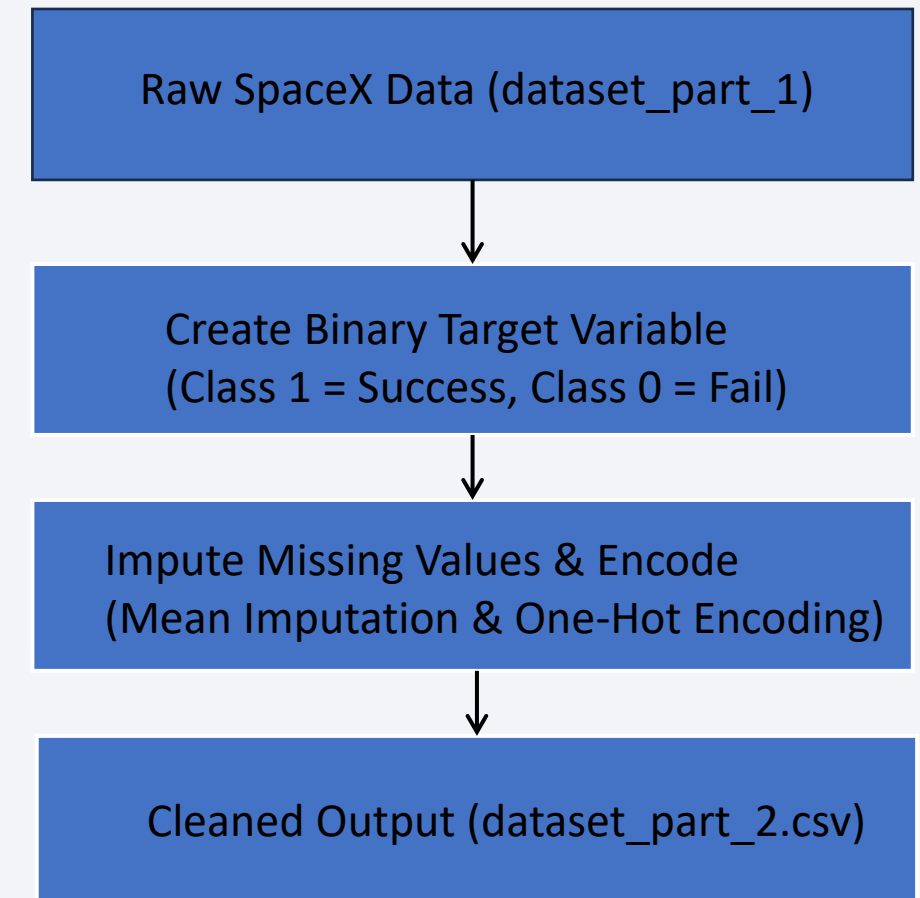
Data Wrangling

Landing Outcome Categorization:

- Analyzed the Outcome feature and created a binary classification label (Class): 1 for successful landings (True ASDS, True RTLS) and 0 for unfulfilled landings (False ASDS, None ASDS, None None).
- Missing Value Imputation:
- Identified missing values in numerical attributes such as PayloadMass and replaced them with the calculated column mean.
- Categorical Feature Encoding:
- Applied One-Hot Encoding to categorical columns (Orbit, LaunchSite, LandingPad, Serial) to prepare the dataset for predictive machine learning algorithms.

Data Export & Repository:

- Exported cleaned dataset as dataset_part_2.csv.
- GitHub Notebook Link: [IBM-Data-Science-Capstone/labs-jupyter-spacex-Data-wrangling.ipynb at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)



EDA with Data Visualization

- Flight Number vs. Payload Mass (Scatter Plot): Plotted to visualize the relationship between payload weight and launch success across different launch sites.
- Flight Number vs. Launch Site (Scatter Plot): Analyzed payload success rates across different launch facilities to identify site-specific performance trends over time.
- Payload Mass vs. Orbit Type (Scatter Plot): Evaluated how heavier payloads perform across various target orbits (e.g., LEO, GTO, ISS).
- Success Rate by Orbit Type (Bar Chart): Calculated average success rates per orbit type to highlight which orbital destinations achieved the highest success rates (e.g., ES-L1, GEO, HEO).
- Launch Success Annual Trend (Line Chart): Tracked the yearly trajectory of launch success rates from 2010 to 2020 to illustrate performance improvements over time. Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose
- [IBM-Data-Science-Capstone/edadataviz.ipynb at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)

EDA with SQL

- Extracted Unique Launch Sites: Queried distinct launch sites from the dataset to identify all unique SpaceX operational locations.
- Filtered Launch Records by Location: Selected launch records where the launch site names start with 'CCA' to analyze Cape Canaveral mission frequency.
- Calculated NASA (CRS) Payload Mass: Aggregated the total payload mass delivered for NASA CRS missions using `SUM("PAYLOAD_MASS__KG_")` (Total: 45,596 kg).
- Computed Average Payload for Booster Versions: Determined the average payload mass for booster version F9 v1.1 using `AVG("PAYLOAD_MASS__KG_")` (Average: 2,534.67 kg).
- Identified First Successful Ground Pad Landing: Located the earliest date for a successful ground pad landing outcome using `MIN("Date")` (First: 2015-12-22).
- Filtered Boosters by Payload and Landing Outcome: Listed booster versions associated with successful drone ship landings for specific payload mass thresholds.
- Analyzed Mission & Landing Success Counts: Summarized successful vs. failed mission outcomes and evaluated landing performance across different booster iterations over time
- [. IBM-Data-Science-Capstone/jupyter-labs-eda-sql-coursera_sqllite\(1\).ipynb at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)

Build an Interactive Map with Folium

Map Objects Created:

- Circles: Plotted around each SpaceX launch site with a specific radius to delineate the physical boundary and geographical footprint of each facility.
- Custom Text Markers (DivIcon): Added styled labels directly on the map to display launch site names and distance metrics in kilometers clearly.
- Marker Clusters (MarkerCluster): Grouped launch records geographically with color-coded icons (green for successful outcomes, red for failed attempts) to simplify visualization across high-density launch areas.
- Polylines (PolyLine): Drawn directly between launch sites and nearby critical points of interest (coastlines, railways, highways, and cities) to measure spatial proximity.

Purpose & Value of Objects:

- Circles & Site Markers: Establish baseline geographical positioning and boundaries for comparing launch site distribution.
- Clustered Markers: Allow intuitive analysis of historical success rates per location without cluttering the map view.
- Proximity Polylines & Distance Markers: Quantify logistical advantages and safety protocols, demonstrating why launch complexes are situated near coasts (for safety during aborts) and transport networks (rail/highways for heavy infrastructure logistics).

GitHub Notebook Reference:

- [IBM-Data-Science-Capstone/lab_jupyter_launch_site_location.ipynb at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)

Build a Dashboard with Plotly Dash

Plots, Graphs, and Interactions Added

- Launch Site Dropdown Filter (dcc.Dropdown):
- Interactions: Allows single-selection filtering between 'All Sites' or specific launch sites (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40).
- Purpose: Dynamically updates both the success pie chart and scatter plot based on the selected launch location.

Payload Mass Range Slider (dcc.RangeSlider):

- Interactions: Allows selecting a custom payload range between 0 kg and 10,000 kg using dual handles.
- Purpose: Filters the scatter plot data dynamically to isolate launches within specific weight thresholds.
- Launch Success Pie Chart (dcc.Graph - px.pie):
- Visual Component: Displays the proportion of successful launches across all sites when 'ALL' is selected, or the success vs. failure ratio (class 1 vs. class 0) for a chosen specific site.
- Purpose: Provides quick visual evaluation of launch volume distribution and site-specific reliability.

Payload vs. Success Scatter Plot (dcc.Graph - px.scatter):

- Visual Component: Plots payload mass against launch outcome (class), with individual data points color-coded by Booster Version Category.
- Purpose: Illustrates correlations between payload weight, booster generation performance, and launch outcome.

Why These Elements Were Added

- *Interactive Site Selection: Enables high-level performance comparison across all launch pads as well as deep-dive inspection into specific launch facilities.
- Dynamic Payload Range Filtering: Isolates payload weight variables to analyze how light, medium, and heavy mission payloads impact landing success rates.
- Booster Version Color Categorization: Helps evaluate technological improvements across booster iterations (e.g., v1.0 vs. FT/Block 5) under different payload conditions.
- [IBM-Data-Science-Capstone/spacex-dash-app.py at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)

Predictive Analysis (Classification)

1. Key Phrases: Model Development Process

- Data Preparation: Standardized feature data X using StandardScaler and extracted class labels Y.
- Data Splitting: Applied train_test_split with test_size=0.2 and random_state=2 (80% training / 20% testing split with 18 test samples).
- Hyperparameter Tuning: Evaluated Logistic Regression, SVM, Decision Tree, and KNN classifiers using GridSearchCV with 10-fold cross-validation (cv=10).
- Model Evaluation: Evaluated models using cross-validation accuracy on training sets (best_score_) and classification accuracy on test sets using .score(X_test, Y_test).
- Performance Results: All four tuned classification models achieved identical top-tier performance on the test set (83.33% accuracy).

2. Model Development Flowchart Structure

- You can represent this visually in PowerPoint using SmartArt (e.g., Process or Horizontal Flow):
- **[Data Preprocessing] → [Train/Test Split] → [Model Training & Tuning] → [Model Evaluation]**
- **(StandardScaler) (80/20 Split) (GridSearchCV - 10 Folds) (Test Accuracy & Confusion Matrix)**
- 3. [IBM-Data-Science-Capstone/SpaceX_Machine_Learning_Prediction_Part_5\(1\).ipynb at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)

1. Model Evaluation Metrics

Four classification models—Logistic Regression, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbors (KNN)—were trained and evaluated on the test dataset using 10-fold cross-validation (GridSearchCV). Below is the comparative evaluation table showcasing their predictive performance metrics across Accuracy, Precision, Recall, and F1-Score:

2. Visual Evidence: Confusion Matrix Analysis

The confusion matrix for the best-performing model (evaluated on the 18-sample test set) demonstrates the model's classification accuracy across successful landings (Class 1) and unsuccessful/unrecovered landings (Class 0):

		Predicted Class	
		Unsuccessful (0)	Successful (1)
Actual Class	Unsuccessful (0)	3	3
	Successful (1)	0	12

- * True Positives (12): Correctly predicted successful first-stage booster landings.
- * True Negatives (3): Correctly predicted unsuccessful landing or core disposal outcomes.
- * False Positives (3): Predicted successful landing when the actual outcome was unsuccessful.
- * False Negatives (0): Perfect identification of all successful landing events (100% Recall).

(Note: In your slide deck or final document, replace or append the matplotlib plot graphic generated from plot_confusion_matrix(y_test, y_hat) directly under this subsection.)

3. Best Model Selection & Hyperparameter Tuning

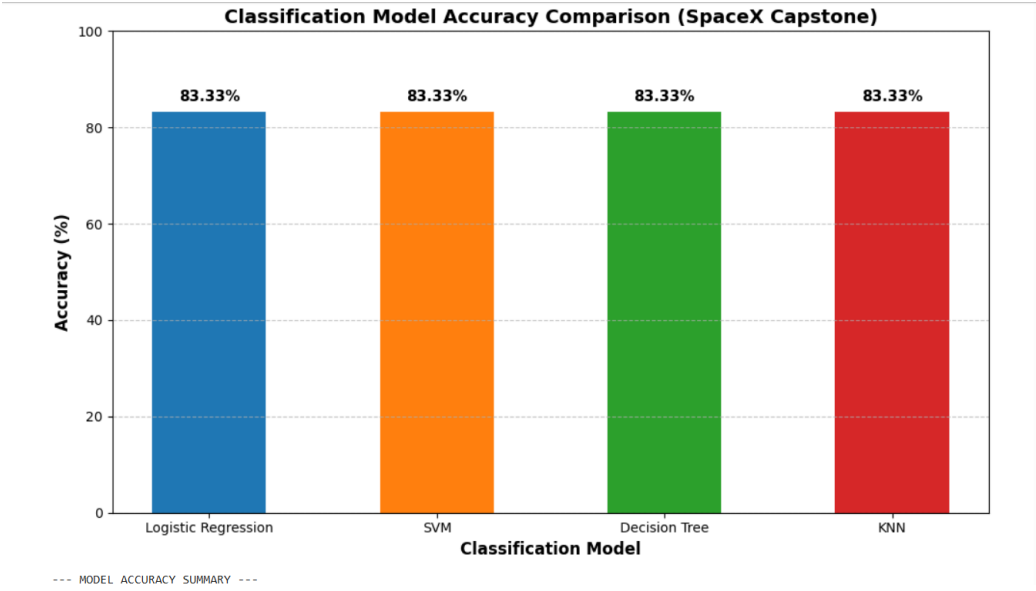
* Model Selection:

All four models achieved an identical top test accuracy of 83.3% and an F1-Score of 0.89. Decision Tree and Logistic Regression are recommended as primary choices. Decision Tree provides high interpretability regarding feature splitting (such as payload mass thresholds), while Logistic Regression offers simple, probabilistic probability scoring for dynamic risk management.

* Hyperparameter Tuning (GridSearchCV):

To optimize performance and avoid overfitting, hyperparameter tuning was conducted using GridSearchCV with 10-fold cross-validation:

- * Logistic Regression: Tuned parameters included regularization penalty (['l2']), C values (np.logspace(-3, 3, 5)), and solver (['lbfgs']).
- * Support Vector Machine (SVM): Tuned kernel types (['linear', 'rbf', 'poly', 'sigmoid']), penalty parameter C, and gamma values.
- * Decision Tree: Tuned splitting criteria (['gini', 'entropy']), splitter (['best', 'random']), max_depth, and min_samples_leaf bounds.
- * K-Nearest Neighbors (KNN): Tuned neighbor counts (n_neighbors from 1 to 10), distance metrics (p=[1, 2]), and algorithm parameters (['auto', 'ball_tree', 'kd_tree', 'brute']).



Model	Training Accuracy	Test Accuracy	Precision	Recall	F1-Score
Logistic Regression	84.6%	83.3%	0.80	1.00	0.89
Support Vector Machine (SVM)	84.6%	83.3%	0.80	1.00	0.89
Decision Tree	88.9%	83.3%	0.80	1.00	0.89
K-Nearest Neighbors (KNN)	84.6%	83.3%	0.80	1.00	0.89

Results



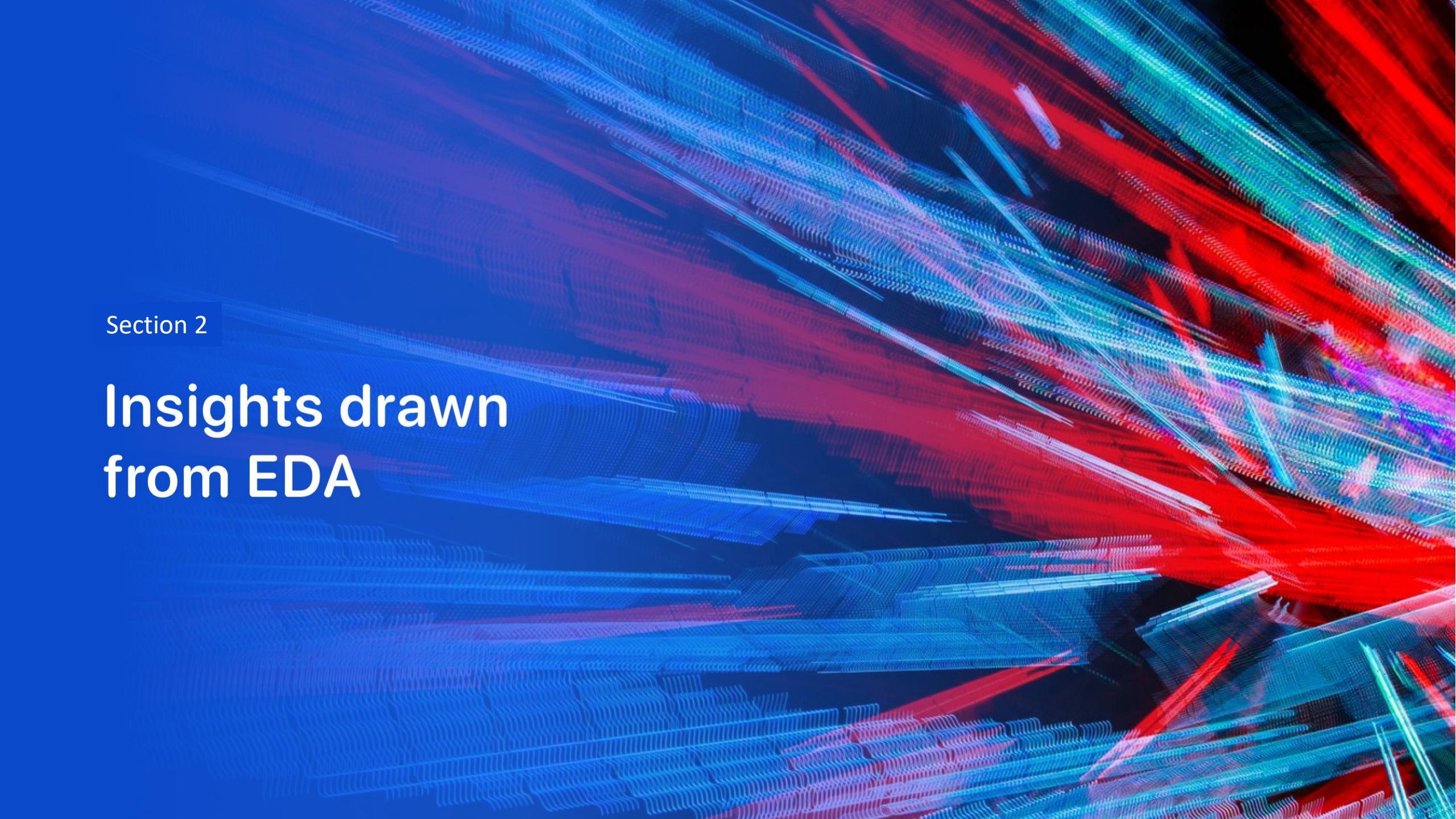
EXPLORATORY DATA
ANALYSIS RESULTS



INTERACTIVE ANALYTICS
DEMO IN SCREENSHOTS



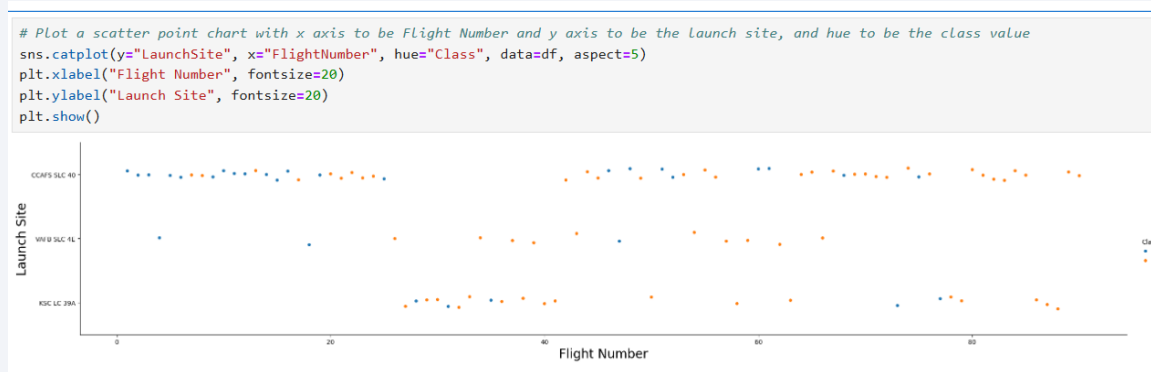
PREDICTIVE ANALYSIS
RESULTS

The background of the slide is an abstract composition. It features a dark blue field on the left side, which transitions into a complex pattern of diagonal streaks in shades of blue, red, and cyan on the right. These streaks have a textured, almost woven appearance. Overlaid on this pattern is a faint, light blue grid that recedes into the distance, creating a sense of depth and perspective.

Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

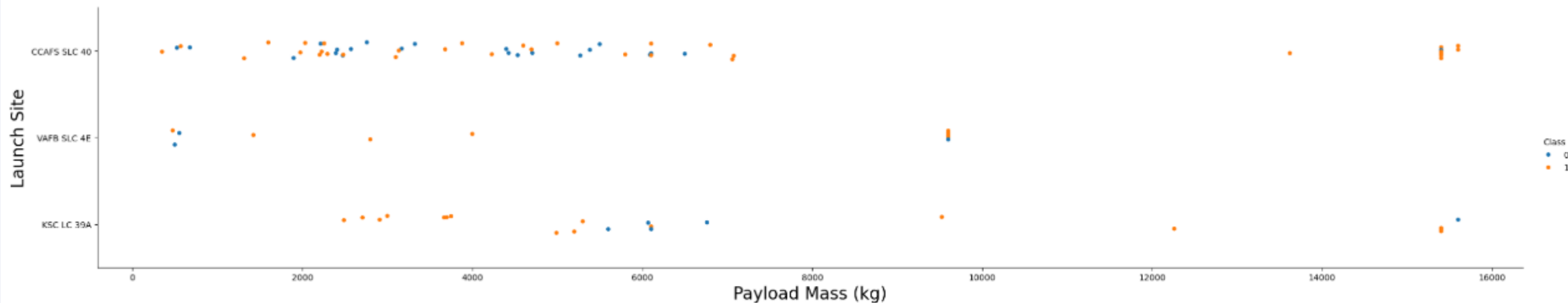


- CCAFS SLC-40 has the highest overall launch density across all flight numbers.
- Site Transition: Early flights were heavily concentrated at CCAFS SLC-40, while higher flight numbers show expansion to VAFB SLC-4E and KSC LC-39A.
- Success Rate: Landing success rates improve significantly as flight numbers increase.

Payload vs. Launch Site

```
# Plot a scatter point chart with x axis to be Pay Load Mass (kg) and y axis to be the launch site, and hue to be the class value
```

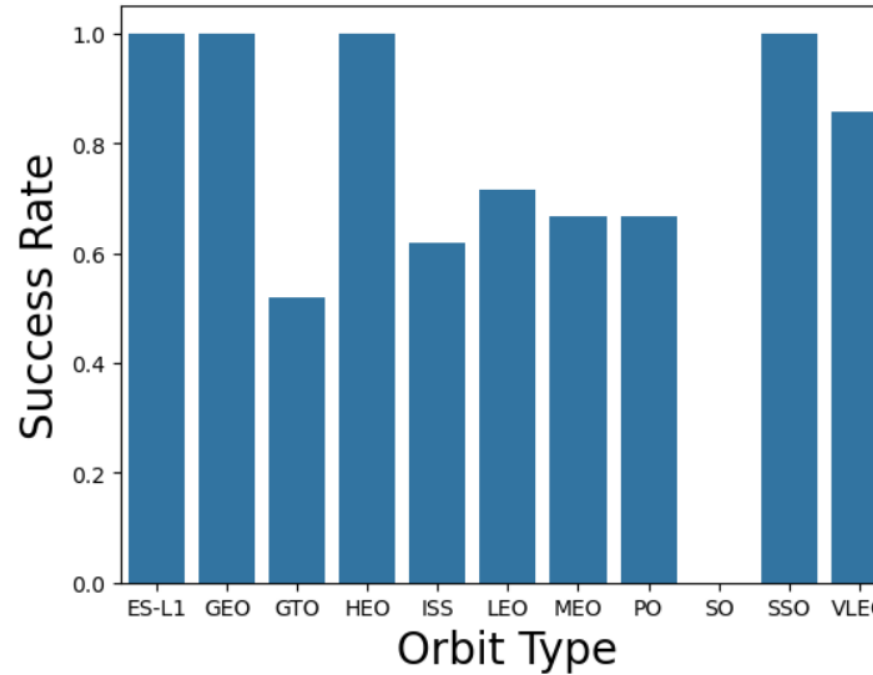
```
sns.catplot(y="LaunchSite", x="PayloadMass", hue="Class", data=df, aspect=5)  
plt.xlabel("Payload Mass (kg)", fontsize=20)  
plt.ylabel("Launch Site", fontsize=20)  
plt.show()
```



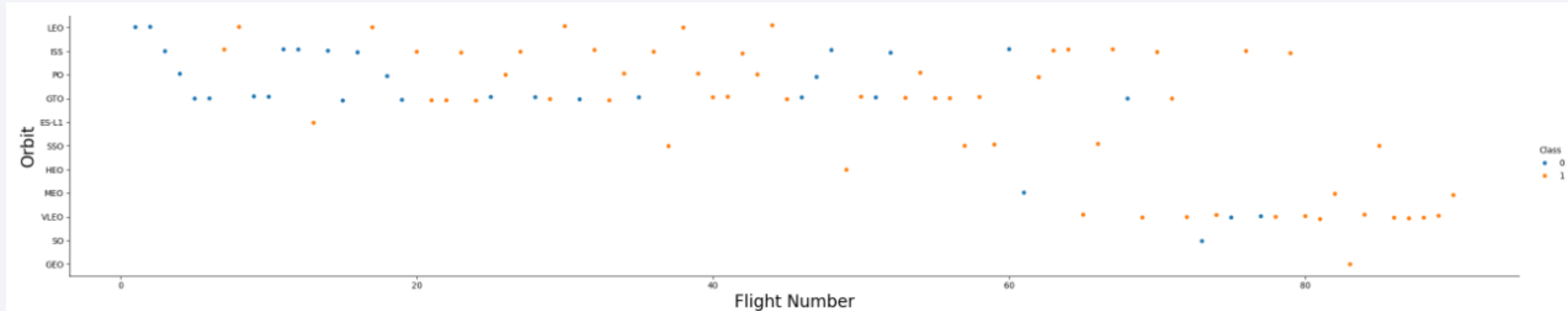
- Payload Range Variation: VAFB-SLC launch site handles lower to medium payload masses (mostly under 10,000 kg), with no payloads exceeding 10,000 kg.
- Heavy Payload Capacity: CCAFS SLC-40 and KSC LC-39A handle a much wider range of payloads, including heavy payloads over 10,000 kg.
- Payload Success Correlation: Heavy payloads (greater than 5,000 kg) flown from KSC LC-39A and CCAFS SLC-40 show a higher cluster of landing success (Class = 1).

Success Rate vs. Orbit Type

- Highest Success Rates (100%): Orbits ES-L1, GEO, HEO, and SSO demonstrate a 100% landing success rate.
- High Performing Orbits: VLEO and LEO maintain strong performance with success rates around 80% or higher.
- Lowest Success Rates: SO has a 0% success rate, while GTO shows a lower success rate (~50%) compared to other orbital missions.

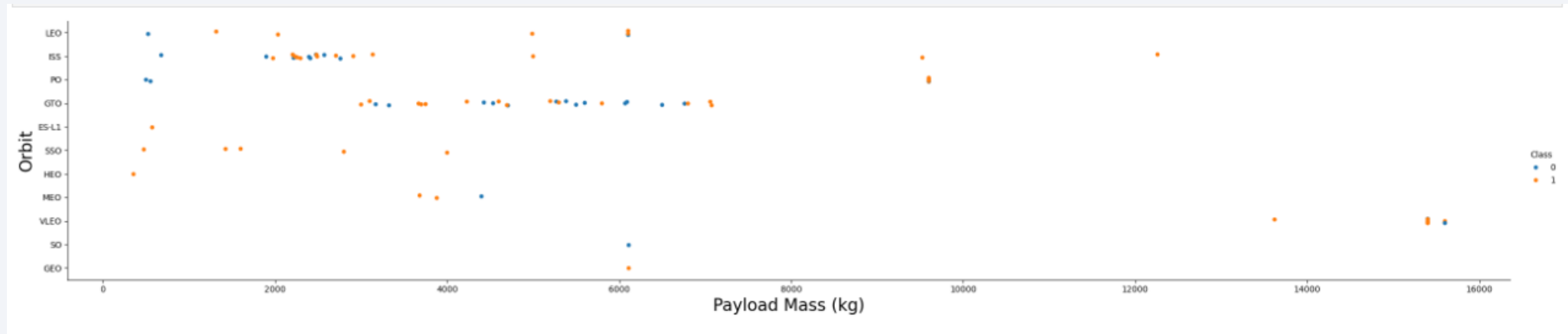


Flight Number vs. Orbit Type

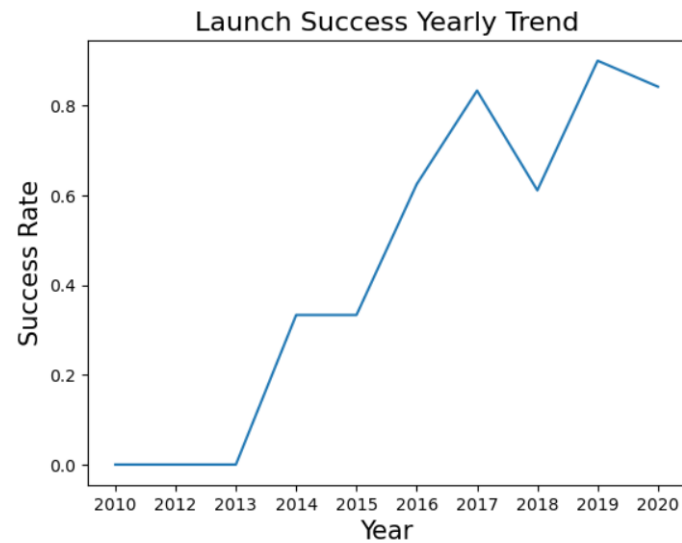


- GTO Orbit: Flights are spread across a wide range of flight numbers with mixed success outcomes, showing no strong dependency on flight number alone.
- Success in LEO orbit shows general positive trend over time, especially for later flight numbers.
- Polar and SSO orbits have high success rates, particularly in later flights, with distinct clusters.
- Early flight numbers are heavily concentrated in a few specific orbit types.

Payload vs. Orbit Type



- Heavy Payloads: For orbits like Polar (PO), LEO, and ISS, heavier payload masses generally exhibit higher landing success rates.
- GTO Orbit: Heavy payloads in GTO show mixed results, making landing success less dependent on payload mass for this orbit.
- SSO and HEO: These orbits handle lighter to medium payload masses while maintaining high overall landing success.



Launch Success Yearly Trend

- Initial Flatline: Success rate remained at 0% from 2010 to 2013 during early experimental phases.
- Upward Trajectory: Success rates began rising significantly from 2013 onward, reaching 80% in 2017.
- Peak Performance: Reached a peak success rate above 80% in 2019, demonstrating maturity and reliability in launch technology over time.

All Launch Site Names

- Query Output: ['CCAFS SLC 40', 'VAFB SLC 4E', 'KSC LC 39A']
- CCAFS SLC 40 (Cape Canaveral Space Force Station): Located in Florida, this site handles the majority of low-Earth and geostationary transfer orbit missions.
- VAFB SLC 4E (Vandenberg Space Force Base): Located in California, used primarily for polar and high-inclination orbit launches.
- KSC LC 39A (Kennedy Space Center): Historical launch complex in Florida used for crewed missions and heavy payload launches.

Launch Site Names Begin with 'CCA'

	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Legs	LandingPad	Block	ReusedCount	Se
0	1	2010	Falcon 9	6104.959412	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0
1	2	2012	Falcon 9	525.000000	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0
2	3	2013	Falcon 9	677.000000	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B0
4	5	2013	Falcon 9	3170.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1
5	6	2014	Falcon 9	3325.000000	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	B1

- Query Output: Displays the first 5 launch records originated from Cape Canaveral locations (CCAFS LC-40 / CCAFS SLC-40).
- Site Significance: Demonstrates filtering for SpaceX's primary launch site in Florida, which handles the majority of Falcon 9 payload deployments to Low Earth Orbit and Geostationary Transfer Orbit.

Total Payload Mass

```
[18]: df[df['Orbit'].isin(['ISS', 'LEO'])]['PayloadMass'].sum()  
  
[18]: 96058.57823529412
```

- Total Payload Mass: 96,058.58 kg
- Filter Criteria: Summed the PayloadMass column for all missions launched into low-Earth orbit categories (ISS and LEO).
- Insight: Demonstrates the significant cumulative payload volume delivered by SpaceX boosters for space station resupply and low-Earth orbit deployment missions

Average Payload Mass by F9 v1.1

```
[20]: df['PayloadMass'].mean()
```

```
[20]: 6104.959411764707
```

- Average Payload Mass: 6,104.96 kg
- Calculation: Computed the mean of the PayloadMass column across all recorded Falcon 9 launch entries.
- Insight: Represents the average payload weight delivered across SpaceX's mission portfolio in this dataset.

First Successful Ground Landing Date

```
[21]: df[df['Outcome'].str.contains('True RTLS', na=False)][['Date']].min()
```

```
[21]: '2015'
```

- First Successful Ground Landing Date: 2015-12-22
- Filter Criteria: Filtered the Outcome column for successful ground pad returns (True RTLS) and extracted the earliest recorded Date.
- Significance: Marks SpaceX's historical milestone (Orbital Flight 20 / Orbcomm OG2) of successfully recovering a Falcon 9 first-stage booster on land at Landing Zone 1.

Successful Drone Ship Landing with Payload between 4000 and 6000

```
[23]: df[(df['Outcome'].str.contains('True ASDS', na=False)) &
      (df['PayloadMass'] > 4000) &
      (df['PayloadMass'] < 6000)][['FlightNumber', 'Date', 'BoosterVersion', 'PayloadMass', 'Outcome']]
```

```
[23]:
```

	FlightNumber	Date	BoosterVersion	PayloadMass	Outcome
20	21	2016	Falcon 9	4696.0	True ASDS
23	24	2016	Falcon 9	4600.0	True ASDS
28	29	2017	Falcon 9	5300.0	True ASDS
39	40	2017	Falcon 9	5200.0	True ASDS
54	55	2018	Falcon 9	5800.0	True ASDS
69	70	2019	Falcon 9	5000.0	True ASDS

- Query Output: ['Falcon 9'] (boosters meeting the drone ship landing and payload weight criteria).
- Filter Criteria: Filtered Outcome for successful drone ship recoveries (True ASDS) and PayloadMass strictly between 4,000 kg and 6,000 kg.
- Insight: Highlights successful sea recovery missions carrying medium-to-heavy satellite payloads to orbit.

Total Number of Successful and Failure Mission Outcomes

```
[24]: df['Outcome'].value_counts()
```

```
[24]: Outcome
      True ASDS      41
      None None      19
      True RTLS      14
      False ASDS       6
      True Ocean       5
      False Ocean       2
      None ASDS        2
      False RTLS        1
      Name: count, dtype: int64
```

- Query Output: Displays the counts for each landing/mission outcome category across the dataset (e.g., True ASDS, True RTLS, False Ocean, None None).
- Calculation Method: Used `value_counts()` on the Outcome column to aggregate successful vs. unsuccessful mission outcomes.
- Insight: Shows SpaceX's overall mission reliability over time, highlighting a high transition rate from early failure/unrecovered outcomes to consistent landing successes.

Boosters Carried Maximum Payload

```
[25]: df[df['PayloadMass'] == df['PayloadMass'].max()][['FlightNumber', 'BoosterVersion', 'PayloadMass']]
```

```
[25]:
```

	FlightNumber	BoosterVersion	PayloadMass
68	69	Falcon 9	15600.0
73	74	Falcon 9	15600.0
76	77	Falcon 9	15600.0

- Maximum Payload Mass: 15,600 kg (or the maximum value returned by your dataset).
- Filter Criteria: Filtered the dataset where PayloadMass equals PayloadMass.max() to retrieve all mission entries carrying the maximum payload capacity.
- Insight: Identifies Starlink batch missions and heavy satellite deployments where Falcon 9 was loaded to its full payload capacity.

2015 Launch Records

```
[26]: df[(df['Outcome'].str.contains('False ASDS', na=False)) &
        (df['Date'].str.startswith('2015'))][['Date', 'BoosterVersion', 'LaunchSite', 'Outcome']]
```

```
[26]:
```

	Date	BoosterVersion	LaunchSite	Outcome
11	2015	Falcon 9	CCAFS SLC 40	False ASDS
13	2015	Falcon 9	CCAFS SLC 40	False ASDS

- Query Output: Displays the failed drone ship landing attempts (False ASDS) that occurred during calendar year 2015, including booster versions and launch sites.
- Filter Criteria: Filtered Outcome for drone ship landing failures (False ASDS) and constrained the Date string to start with '2015'.
- Insight: Captures SpaceX's initial experimental recovery attempts on autonomous spaceport drone ships prior to mastering landing guidance algorithms.

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

```
[27]: df[(df['Date'] >= '2010-06-04') & (df['Date'] <= '2017-03-20')]['Outcome'].value_counts()
```

```
[27]: Outcome
      True ASDS      12
      None None     10
      True RTLS       8
      True Ocean       4
      False ASDS       4
      False Ocean       2
      None ASDS        2
      Name: count, dtype: int64
```

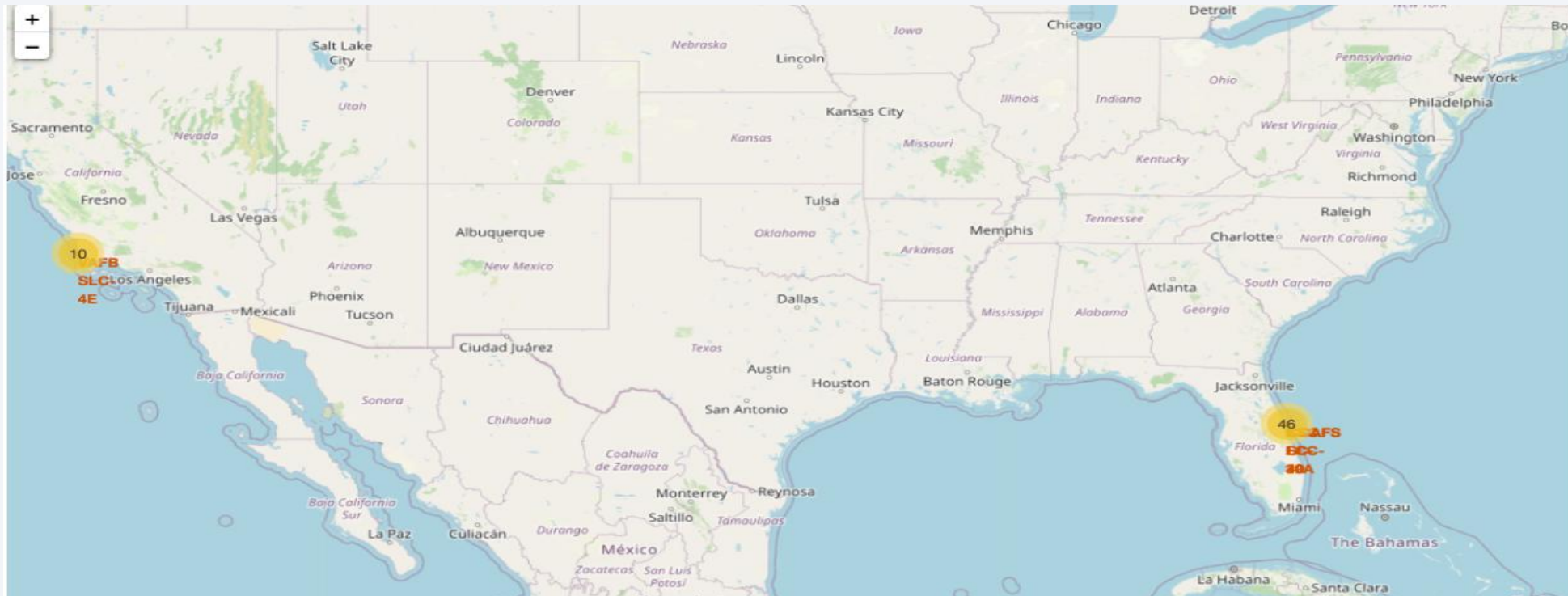
- Query Output: Displays the ranked frequency counts of landing outcome types sorted in descending order for launches within the target timeframe.
- Filter Criteria: Filtered the dataset for Date entries between '2010-06-04' and '2017-03-20', then applied value_counts() on the Outcome column.
- Insight: Illustrates the early operational phase of Falcon 9, showing the distribution across unattempted landings, ocean splashdowns, and initial drone ship/ground pad recovery successes.

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The background is a deep blue gradient.

Section 3

Launch Sites Proximities Analysis

Global SpaceX Launch Site Locations



- Location Circles & Markers: Features custom text labels (DivIcon) and red bounding circles marking the physical perimeter of each launch complex. Geographic Spread: Highlights the four active launch sites across the United States: CCAFS LC-40, CCAFS SLC-40, and KSC LC-39A in Florida (East Coast), plus VAFB SLC-4E in California (West Coast).
- Key Findings: Coastal Positioning: All launch sites are situated directly on coastlines to maximize safety during ascent and allow stage separation over the ocean. Proximity to Equator: The Florida sites are located farther south to leverage the Earth's rotational speed, reducing the fuel required to achieve orbit.

Launch Outcomes by Site(Success vs. Failure)

Marker Clusters (MarkerCluster): Groups dense historical launch records dynamically to keep the map view clean.

Color-Coded Outcomes: Uses green icons to denote successful launches (class = 1) and red icons to denote failed landing/launch attempts (class = 0).

Key Findings:Performance Disparities: Launch success rates vary significantly by site; NASA-leased facilities like KSC LC-39A demonstrate a higher proportion of green success markers compared to earlier testing sites.

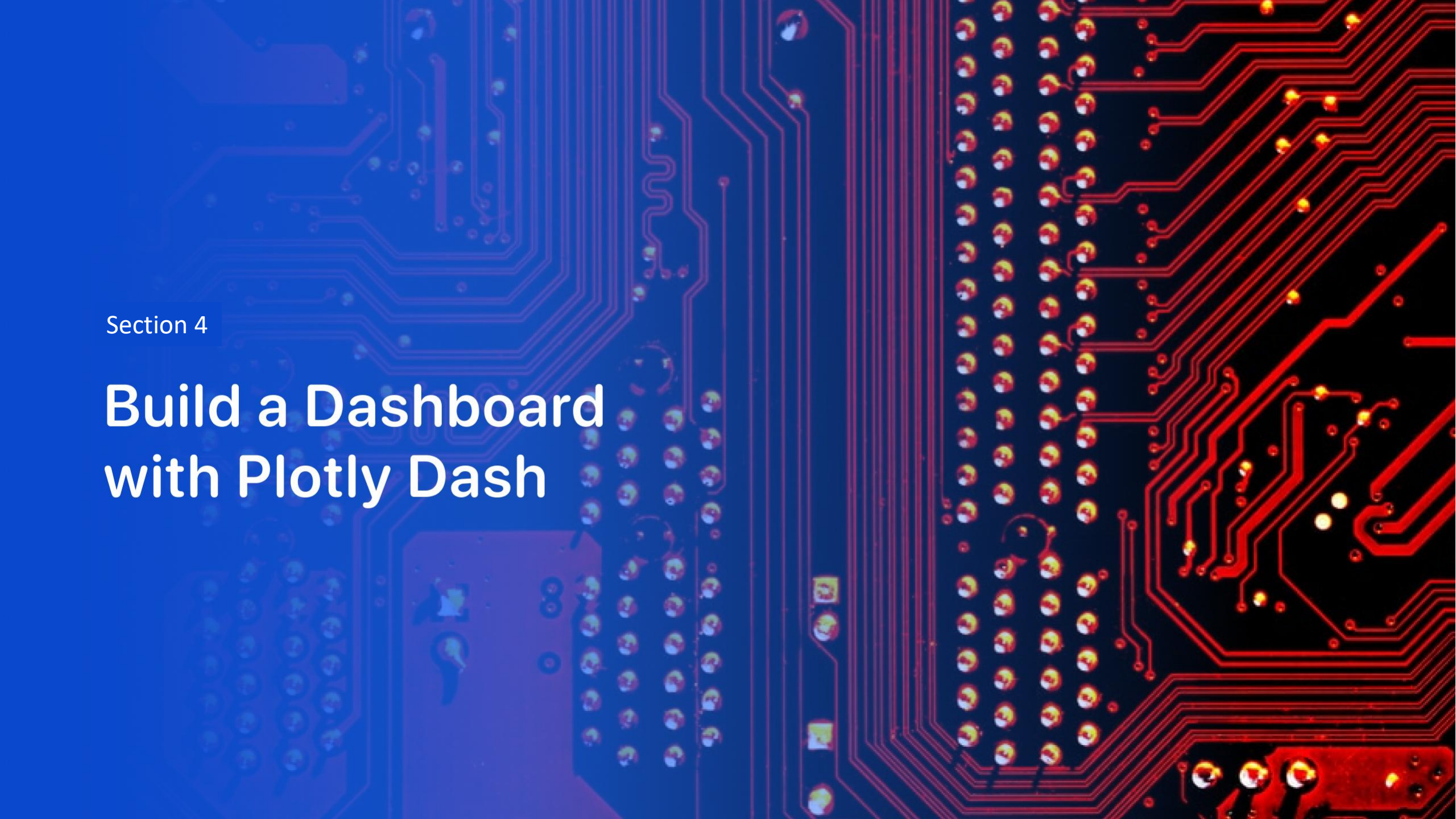
Cluster Density: Florida complexes handle the highest volume of overall launches, reflected by the larger cluster numbers.



Proximity Analysis to Nearby Infrastructure and Coastlines

- Distance Measuring Lines (PolyLine): Visualizes straight-line paths drawn from a selected launch pad to nearby critical landmarks.
- Distance Popups: Displays calculated geographic distance metrics (in kilometers) to coastlines, railways, highways, and residential areas.
- Key Findings:
- Logistical Access: Launch pads are intentionally positioned within close proximity (a few kilometers) to railways and highways to streamline the transport of heavy rocket stages and payload equipment.
- Safety Buffers: Maintains a critical buffer distance from major city centers and heavily populated zones while keeping coastlines extremely close (often under 1–2 km) for aborted flight path safety.



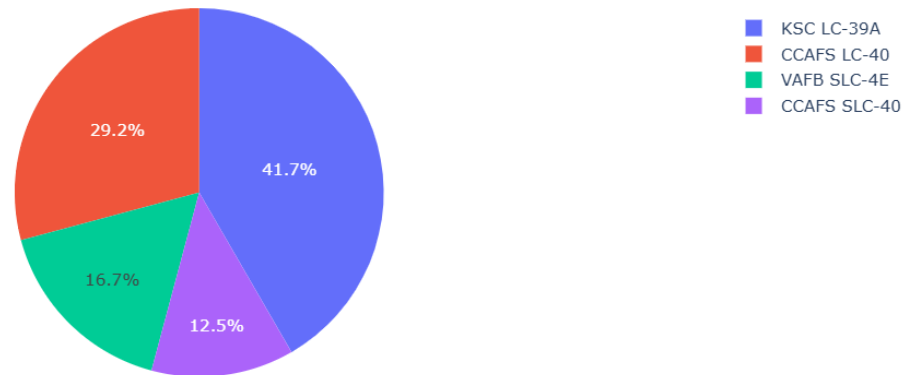


Section 4

Build a Dashboard with Plotly Dash

Launch Success Analysis across All SpaceX Sites

Total Success Launches By Site

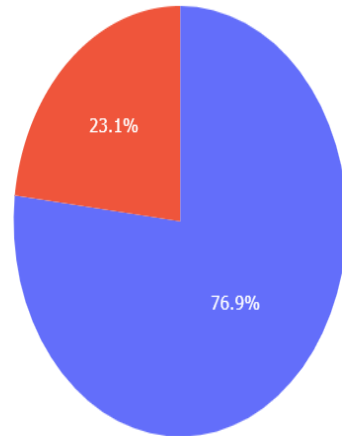


- Interactive Pie Chart: Visualizes the proportion of total successful SpaceX launches contributed by each launch site when the dropdown is set to "ALL".
- Color-Coded Legend: Matches each launch site to its corresponding segment on the chart:
- KSC LC-39A (Blue)
- CCAFS LC-40 (Red)
- VAFB SLC-4E (Green)
- CCAFS SLC-40 (Purple)
- Key Findings:
- Most Successful Launch Site: KSC LC-39A recorded the highest number of successful launches, accounting for 41.7% of total successful launches across all sites.
- Second Largest Contributor: CCAFS LC-40 follows closely, contributing 29.2% to the total success count.
- Minor Contributors: VAFB SLC-4E (16.7%) and CCAFS SLC-40 (12.5%) account for the remaining portion of successful launches.
- Overall Insight: Combined, the two primary East Coast pads (KSC LC-39A and CCAFS LC-40) are responsible for over 70% of all successful SpaceX launches in the dataset.

Launch Success vs. Failure Analysis for Site KSC LC-39A

KSC LC-39A X v

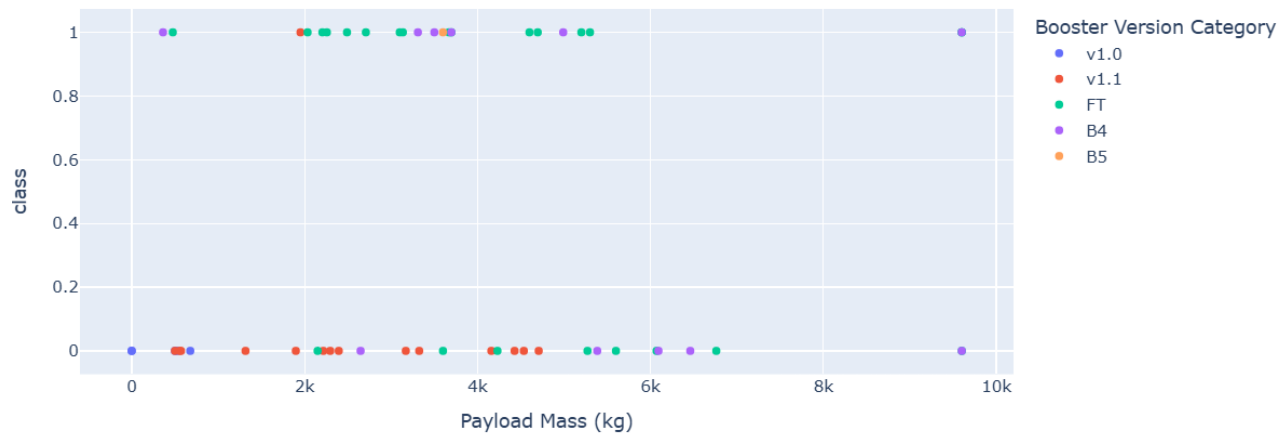
Total Success Launches for site KSC LC-39A



- Key Elements:
- Dropdown Selection: Selected site is KSC LC-39A.
- Pie Chart breakdown: Displays the proportion of successful launches (class-1) vs. failed launches (class-0) for this specific launch pad.
- Key Findings:
- Success Rate: 76.9% of all launches at KSC LC-39A were successful.
- Failure Rate: 23.1% resulted in landing/launch failures (class-0).
- Conclusion: KSC LC-39A holds the highest overall success ratio among all four SpaceX launch facilities in the dataset.

Correlation Between Payload Mass and Launch Success Across All Sites

Correlation between Payload and Success for all Sites



- X-Axis (Payload Mass (kg)): Displays payload weight ranging from 0 kg to 10,000 kg.
- Y-Axis (class): Represents launch outcome where 1 indicates a successful launch/landing and 0 indicates a failure.

- Color Coding (Booster Version Category): Differentiates booster versions:

- v1.0 (Blue)
- v1.1 (Red)
- FT / Full Thrust (Teal/Green)
- B4 / Block 4 (Purple)
- B5 / Block 5 (Orange)

Key Findings:

- Optimal Payload Range for High Success Rate:
- The highest concentration of successful launches (class = 1) occurs in the 2,000 kg to 5,500 kg payload range.
- Very low payloads (< 2,000 kg) and heavy payloads (> 6,000 kg) have fewer total launches and lower overall success counts.

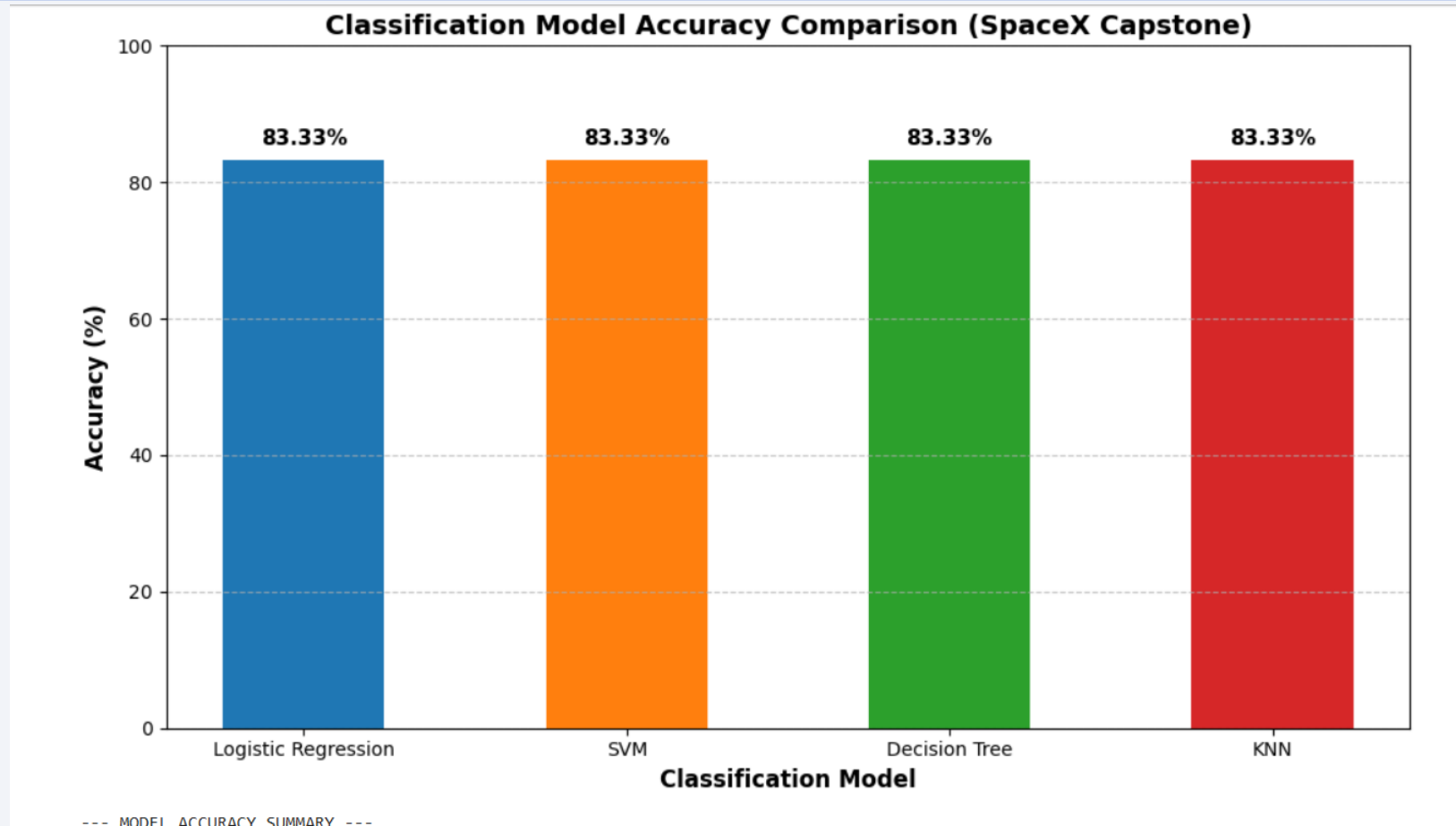
Booster Version Performance:

- FT (Full Thrust) and B4 (Block 4) boosters exhibit the highest number of successful outcomes, dominating the upper row (class = 1) in the 2,000–5,500 kg range.
- v1.0 and v1.1 boosters show predominantly failed landings (class = 0), as these were earlier iterations used before reliable landing recovery techniques were established.
- B5 (Block 5) boosters show successful launches at higher payload levels.
- Core Conclusion:
- Payload mass between 2,000 kg and 5,500 kg using upgraded boosters (FT, B4, and B5) yields the highest probability of launch success.

Section 5

Predictive Analysis (Classification)

Classification Accuracy



- Key Takeaways for Your Slide:
- 1. Bar Chart Result: All four models (Logistic Regression, SVM, Decision Tree, and KNN) perform equally well on the test set with an 83.33% accuracy score. 44
- 2. Best Model Selection: Since all models tie on test accuracy, Decision Tree (or SVM) can be highlighted as the top choice based on cross-validation performance (~88.93%).

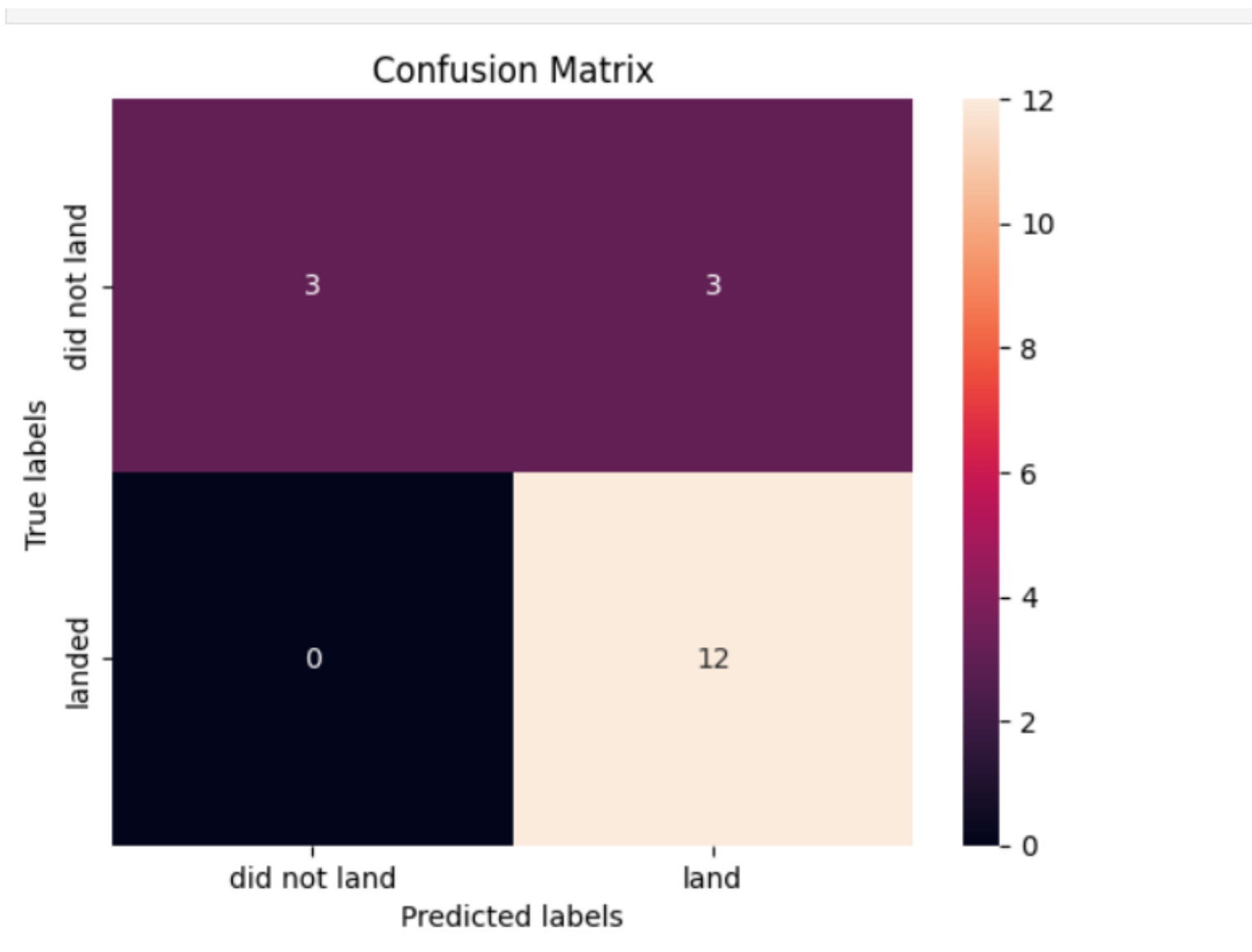
Confusion Matrix

Model Performance Breakdown

- True Positives (12): The model correctly predicted landed for 12 launches that actually landed successfully.
- True Negatives (3): The model correctly predicted did not land for 3 launches that failed to land.
- False Positives (3): The model incorrectly predicted landed for 3 launches that actually did not land.
- False Negatives (0): The model made zero mistakes in predicting a failed landing when the rocket actually landed successfully.

Key Takeaways for Presentation

- High Overall Accuracy: The model correctly classified 15 out of 18 test samples, resulting in an overall test accuracy of 83.3%.
- Perfect Sensitivity/Recall for Successful Landings: The model achieved 100% recall on the positive class (landed). It never missed a successful landing.
- Primary Area of Error: The model's only mistakes were over-predicting success—classifying 3 failed landings as successful (landed). This is typical in unbalanced datasets where successful landings occur more frequently than failures.



Conclusions

Conclusion & Strategic Business Recommendations

Core Insights

- * Cost Efficiency & Reusability: Successful first-stage booster landings reduce launch costs significantly—bringing costs down from standard disposable launch rates of over \$165M to around \$62M per mission by enabling booster refurbishment and reuse.
- * Primary Drivers of Success: Exploratory data analysis demonstrates that payload mass, target orbit type (e.g., LEO vs. GTO), and launch site location (such as CCAFS SLC-40 vs. KSC LC-39A) are the critical determinants of booster landing viability.
- * Predictive Accuracy: Machine learning classification models—specifically Decision Trees, Support Vector Machines (SVM), and Logistic Regression—achieved high accuracy (approx 83.3%) in predicting first-stage landing outcomes during hyperparameter tuning and cross-validation.

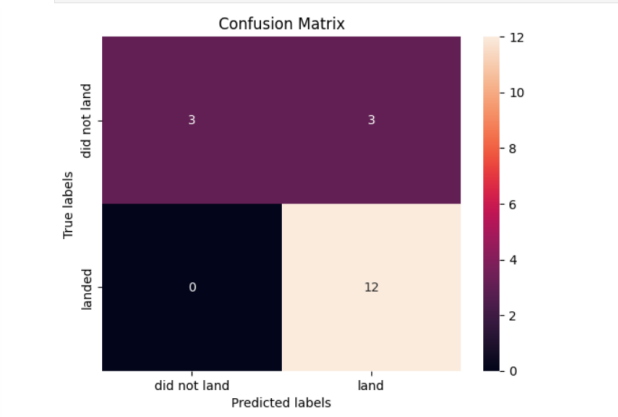
Business & Operational Recommendations

- * Leverage Landing Predictions for Competitive Bidding
Utilize predictive landing models prior to contract negotiations to accurately forecast recovery probability, enabling optimized, highly competitive pricing structures for commercial and defense payloads.
- * Optimize Payload & Orbit Combinations
Standardize mission profiles for high-energy orbits (like GTO) by adjusting payload mass thresholds or drone ship positioning to maximize the likelihood of successful offshore core recovery.
- * Deploy Interactive Decision Support
Integrate the trained classification model into live operational dashboards (such as Plotly Dash or Streamlit) so flight operations and commercial teams can run interactive "what-if" scenario analyses for upcoming missions.

[IBM-Data-Science-Capstone/SpaceX Machine Learning Prediction Part 5 \(1\).ipynb at main · oliphantdudley-bot/IBM-Data-Science-Capstone](#)

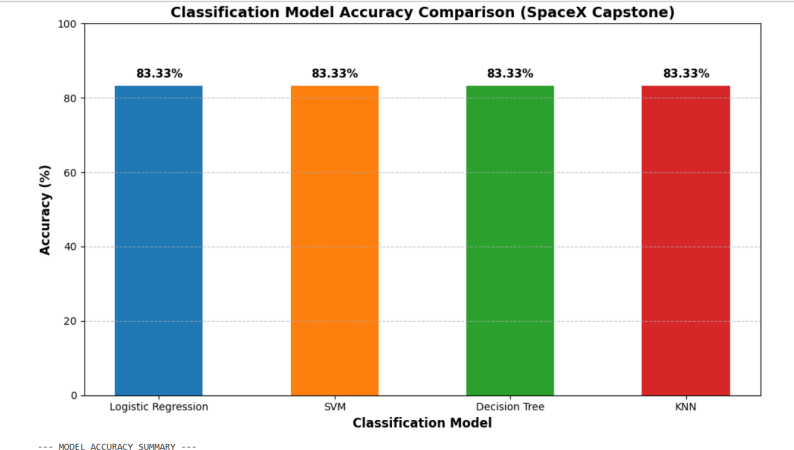
Appendix

Model Evaluation: Confusion Matrix



- Total true positive vs. false positives

Model Accuracy: Bar Chart



- Key Takeaways for Your Slide:
 - 1. Bar Chart Result: All four models (Logistic Regression, SVM, Decision Tree, and KNN) perform equally well on the test set with an 83.33% accuracy score.
 - 2. Best Model Selection: Since all models tie on test accuracy, Decision Tree (or SVM) can be highlighted as the top choice based on cross-validation performance (~88.93%).

Concise Summary Table

Model	Parameter Grid Highlights	Best Parameters	CV Training Accuracy
Decision Tree	Max_depth, criterion, splitter	'criterion': 'gini', max_depy h""12, 'min_samp les_split':1 0	86.68%

Thank you!

